

OG-PHL Examples: From 5-day Workshop

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Examples

- (DoF) Simulation of Capital Gains Tax and VAT reform using OG-PHL
- (Ateneo) Macroeconomic Consequences of Disasters in the Philippines: OG-PHL Simulations
- (NEDA) Impact of Digitization on the Philippine Economy
- (DBM) Accelerating Government Infrastructure Investments: Boon or Bane?

From 5-day Workshop: August 2024

“Simulation of Capital Gains Tax and VAT reform using OG-PHL”

Adrian Castro, Alexandra Leonardo, Aris Zoleta, Gabriel Salar, Gladys Audar, Lance Garcia, Justin Simon, Sarah Conche

- Had a great calibration of β_j 's to match wealth moments
- Used (external) data from the World Inequality Database (<https://wid.world/>) to calibrate the share of wealth held by each percentile group.

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Percentile	β_j	Model wealth share	Data wealth share
Top 1%	0.99	32%	32.26%
90-99th%	0.98	31%	31.25%
80-90th%	0.80	14%	14.07%
70-80th%	0.70	8%	8.62%
50-70th%	0.50	7%	9.71%

- Based on steady-state wealth shares

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“Macroeconomic Consequences of Disasters in the Philippines: OG-PHL Simulations”

Philip Arnold P. Tuano, Gay D. Defiesta, Jhon Louie B. Sabal, Christian C. Pasion, Joe Mari S. Francisco, Paolo Magnata, Cymon Kayle Lubangco

- What is the effect of a 100-year typhoon shock on the economy?
- Looked at earnings, disutility of labor, and TFP shocks

Baseline + 3 scenarios

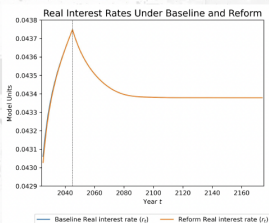
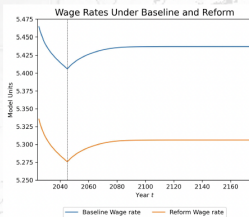
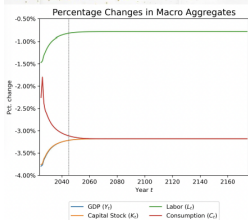
Scenario 1	Scenario 2	Scenario 3	Scenario 4
Baseline	Decrease in deterministic ability process (e) by 1% Increase in disutility of labor (χ_n) by 1% No change in total factor productivity (Z)	Decrease in deterministic ability process (e) by 1% Increase in disutility of labor (χ_n) by 1% Decrease in total factor productivity (Z) by 1%	Decrease in deterministic ability process (e) by 1% Increase in disutility of labor (χ_n) by 1% Increase in total factor productivity (Z) by 1%

Changes in code

Scenario 2	Scenario 3	Scenario 4
#reduced earning ability $p2.e = p.e * 0.99$	#reduced earning ability $p2.e = p.e * 0.99$	#reduced earning ability $p2.e = p.e * 0.99$
#increased disutility of labor $p2.chi_n = p.chi_n * 1.01$	#increased disutility of labor $p2.chi_n = p.chi_n * 1.01$	#increased disutility of labor $p2.chi_n = p.chi_n * 1.01$
#no change in TFP	#decreased in TFP $p2.Z = p2.Z * 0.99$	#increased in TFP $p2.Z = p2.Z * 1.01$

Scenario 3 output

Scenario 3: Wages are lower if TFP declines



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“Impact of Digitization on the Philippine Economy”

Roxanne Elumbre, Kristine Magaling, Clarisa Manzon, Dorothy Obispo

- In the Philippines, the prices of fixed broadband plan is 19.6 percent higher compared with the average price in other ASEAN countries.
- Increase productivity of the bottom deciles and reduce capital depreciation

From 5-day Workshop: August 2024

“Accelerating Government Infrastructure Investments: Boon or Bane?”

Arielle Fajardo, FrabertDela Cruz

- “Build-Better-More” Program
- Increase government investment spending by 5% of GDP
- Financing via debt vs. tax increases